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The Adler Fractal

A Recursive Radial Drawing and Its Polygon-IFS Limits

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Abstract

This report documents a recursive radial drawing that I first wrote as a short Python Turtle script and that I call the *Adler Fractal*. Every node draws equally spaced spokes in the same absolute directions; the next generation starts at their endpoints. The original uses 72 directions, two generations and spokes of constant length. The report separates the accumulated trace, the endpoint sets and their closures, and derives command counts, endpoint multiplicities, sharp radius bounds and the tangent circles that form the rings of the drawing. For the equal-length original, the union of all endpoint generations is dense in the plane. Contractive variants produce established regular polygon iterated function systems; their trace closures are established inhomogeneous self-similar sets. A Python implementation without GUI dependencies, a comparison with the original script, proofs and reproducible figures accompany the report. I claim neither a new fractal family nor historical priority.

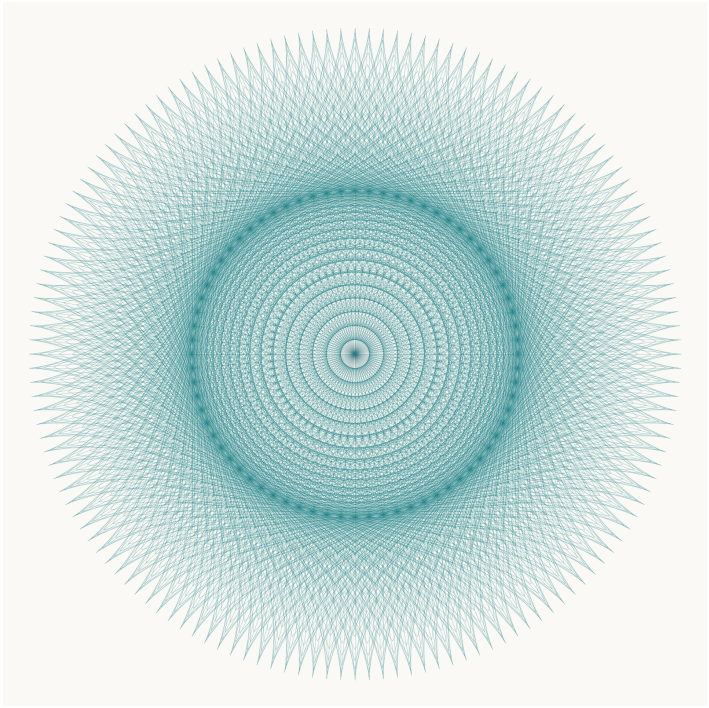


Figure 1: Original geometry: $m = 72$, $n = 2$, $L = 200$, $r = 1$. The 5,256 drawing commands create a tightly packed radial rosette. Color and stroke width are rendering choices, not part of the geometric definition.

Provenance. My original script is included as `original/eagle_fractal.py`; Section 1 notes one correction. Formalization, analysis, text and reference software were prepared with AI assistance (OpenAI ChatGPT; revised with Anthropic Claude). There has been no independent mathematical review.

1. Definition and relation to the original script

Identify the Euclidean plane with $\mathbb{C}$. Fix an integer $m \geq 3$, a depth $n \geq 0$, an initial length $L > 0$, a scale $0 < r \leq 1$, an orientation $\theta \in \mathbb{R}$ and an origin $z_0 \in \mathbb{C}$. Define

$$u_j = e^{i(\theta + 2\pi j/m)}, \qquad j = 0, \ldots, m-1.$$

The same global directions are used at every node. No direction is rotated relative to the incoming segment. For a word $w = (j_1, \ldots, j_k)$, put

$$p_w = z_0 + L \sum_{t=1}^{k} r^{t-1} u_{j_t}, \qquad p_\emptyset = z_0. \tag{1}$$

Let w^- be the word with its last symbol removed. Define the generation-k endpoint set and the accumulated trace by

$$E_k = \{p_w : |w| = k\}, \tag{2}$$

$$A_n = \bigcup_{k=1}^{n} \bigcup_{|w|=k} [p_{w^-}, p_w], \qquad A_0 = \emptyset. \tag{3}$$

Here $[a, b]$ denotes a closed straight segment. An endpoint can have several addresses. Coincident drawing commands count separately in a program, but the geometric union contains each point only once.

For $z_0 = 0$, write

$$B = \bigcup_{j=0}^{m-1} [0, Lu_j], \qquad f_j(x) = Lu_j + rx.$$

For $n \geq 1$ the finite trace satisfies

$$A_n = B \cup \bigcup_{j=0}^{m-1} f_j(A_{n-1}). \tag{4}$$

Other origins translate all sets by z_0. Subsequent sections use origin zero.

Original parameters and traversal

The main call of the original script uses

$$(m, n, L, r, \theta, z_0) = (72, 2, 200, 1, 0, 0), \quad \alpha = 360^\circ/m = 5^\circ.$$

At a node, all m spokes are drawn first in increasing angle order. The children are then visited recursively in the same order. Repositioning between segments happens with the pen raised. The reference generator preserves this ordering, including retraced segments.

The original accepts any integer step in `range(0, 360, ang)`. If the step is below 360 and does not divide it, the cyclic gap at the end differs from the other gaps. Such nonregular direction sets are outside this specification; the angle-import adapter rejects them, as it does the steps 180 and 360, which leave fewer than three directions. The function's default step is 20 degrees, while the main call uses 5 degrees.

In the first published version of the script (v1.0.0), the drawing function received `_turtle` but drew with a module-level global `turt`, so it ran only as a script. Since v1.0.1 it draws with the turtle it is given. This changes the dependency, not the drawing: the same 5,256 segments are produced in the same order.

2. Finite counts, geometry and multiplicity

Proposition 1. *The number of drawing commands and their total length, with multiplicity, are*

$$N_n = \sum_{k=1}^{n} m^k = \frac{m(m^n - 1)}{m - 1}, \qquad T_n = Lm \sum_{k=0}^{n-1} (mr)^k.$$

The sharp outer radius is $R_n = nL$ for $r = 1$ and $R_n = L(1 - r^n)/(1 - r)$ for $0 < r < 1$.

Proof. Generation k has m^k addresses and length Lr^{k-1} per command. Summing gives the first two formulas. The triangle inequality bounds every endpoint by R_n; a word repeating a single direction attains it. The ball of that radius also contains all segments, by convexity. □

Depth n	Commands, $m = 72$	Outer radius, $L = 200$, $r = 1$
1	72	200
2	5,256	400
3	378,504	600
4	27,252,360	800

In particular, $T_2 = 1{,}051{,}200$ length units for the original. This is not the length of the trace after removing overdraw.

Symmetry and finite dimension

Rotation through $2\pi/m$ permutes the direction indices. Reflection in an axis through a direction does likewise. Thus A_n and E_n are invariant under the corresponding dihedral group. For $n \geq 1$, A_n is connected: each new star attaches at an endpoint of an earlier segment. It is a finite union of nondegenerate segments, so $\dim_{\mathrm{H}} A_n = 1$.

Why the second generation looks circular

For $r = 1$ and $n = 2$,

$$|p_{(a,b)}| = L|u_a + u_b| = 2L \left| \cos \frac{\pi(a - b)}{m} \right|.$$

These discrete radii explain the rings in the endpoint pattern. They do not assert that entire circles are present in the trace. The rings that dominate the drawing of the trace inside the circle $|z| = L$ are tangent circles of its segments (Proposition 2 and the remarks after it); near the center they almost coincide with these endpoint radii.

Still with $r = 1$, for even m the second endpoint set has exactly $m^2/2 + 1$ distinct points. Indeed, $p_{(a,b)}/2$ is the midpoint of the chord from Lu_a to Lu_b (a single point of the circle if $a = b$), and a nonzero midpoint determines the unordered pair $\{a, b\}$. There are $m(m + 1)/2$ unordered pairs allowing repetition; all $m/2$ antipodal pairs collapse to zero, and all other pairs remain distinct. Hence the count is $m(m+1)/2 - m/2 + 1$. For $m = 72$, 5,184 endpoint addresses therefore represent only 2,593 points. The origin has 72 ordered addresses.

Tangent circles and chords of the second generation

Keep $r = 1$. For direction indices a, b let $\delta = 2\pi(b - a)/m$ be the angle from u_a to u_b, and let $\sigma_{a,b} = [Lu_a, L(u_a + u_b)]$ be the corresponding second-generation segment. It starts on the *node circle* $|z| = L$, where the spokes of the first generation end.

Proposition 2. *If $\cos\delta \le 0$, the point of $\sigma_{a,b}$ nearest to the origin is $L(u_a - \cos\delta\, u_b)$, at distance $L|\sin\delta|$. Thus $\sigma_{a,b}$ touches the circle $|z| = L|\sin\delta|$ and does not enter the open disk it bounds. If $\cos\delta > 0$, the nearest point is the starting point Lu_a. A chord $[Lu_a, Lu_c]$ of the node circle with $c \not\equiv a \pmod m$ lies in A_2 if and only if m is even and $c - a \equiv m/2 \pmod 2$.*

Proof. For $0 \le t \le 1$,

$$|u_a + tu_b|^2 = 1 + 2t\cos\delta + t^2 = \sin^2\delta + (t + \cos\delta)^2.$$

Over the real line the minimum lies at $t = -\cos\delta$, which belongs to $[0, 1]$ exactly when $\cos\delta \le 0$; otherwise the expression increases on $[0, 1]$. Multiplying by L gives the first two claims.

Now let m be even and $\cos\delta < 0$. The line of $\sigma_{a,b}$ meets the node circle again at $t = -2\cos\delta$, in the point $L(u_a - 2\cos\delta\, u_b) = -Le^{2i\delta}u_a = Lu_c$ with $c = 2b - a - m/2$, an index because m is even. The segment $\sigma_{c,b+m/2}$ starts there and runs back along the same line, since $u_{b+m/2} = -u_b$. Both segments have length L and the chord has length $2L|\cos\delta| \le 2L$, so together they cover it. As $b - a$ runs through the residues with $\cos\delta < 0$, the difference $c - a = 2(b - a) - m/2$ runs through all residues $j \not\equiv 0 \pmod m$ with $j \equiv m/2 \pmod 2$.

Conversely, A_2 is a finite union of segments with directions $\pm u_j$. A nondegenerate segment inside it meets every piece that is not collinear with it in at most one point, so it is covered by collinear pieces and is parallel to some u_j. The chord $[Lu_a, Lu_c]$ has the direction angle $\theta + \pi(a + c)/m + \pi/2$, so it is parallel to u_j exactly when $a + c + m/2 \equiv 2j \pmod m$. For odd m the left side is not an integer. For even m such a j exists exactly when $a + c + m/2$ is even, that is, when $c - a \equiv m/2 \pmod 2$. □

For the original, δ is a multiple of $5°$. Call the m segments $\sigma_{a,a+d}$ with fixed d a *segment class*; the classes d and $-d$ touch the same circle. The classes with $\cos\delta \le 0$ touch the circles of radius $L\sin(5k°)$, $k = 0, \ldots, 18$: the center for $k = 0$, where segments point straight back; the node circle for $k = 18$, which segments touch at their starting points; and 17 circles in between. The identity in the proof shows why these circles are visible. Along a segment, the distance ρ from the origin changes only quadratically near the point of contact, so on each side of that point a segment carries the arc length $\rho/\sqrt{\rho^2 - h^2}$ per unit of radius, where $h > 0$ is the tangent radius; this is unbounded as $\rho \to h$. The m rotated copies in a segment class therefore crowd along their common tangent circle, as threads do in string art. The prominent concentric rings inside the node circle in Figure 1 lie at these radii, the outermost ones merging into the dark band along the node circle; none of them is drawn as a curve.

For $\cos\delta \le 0$, the endpoint of $\sigma_{a,b}$ lies $L(1 + \cos\delta) = 2L\cos^2(\delta/2)$ beyond the point of contact, and the tangent radius $L|\sin\delta|$ is $|\sin(\delta/2)|$ times the endpoint radius $2L|\cos(\delta/2)|$. For δ close to π a segment ends just after touching its circle, so near the center the tangent radii and the endpoint radii above nearly agree: $L\sin 5° \approx 0.0872L$ against $2L\sin 2.5° \approx 0.0872L$. Further out they separate, for example $L\sin 40° \approx 0.643L$ against $2L\sin 20° \approx 0.684L$; in Figure 1 the endpoint radii add at most faint features.

For $m = 72$ the chords in A_2 are exactly those with an even index difference, with central angles $10°, 20°, \ldots, 180°$. At $\delta = \pm 120°$ a single segment already joins two nodes, and these chords form 12 regular hexagons. The second generation thus contains closed regular polygons and star polygons inscribed in the node circle, with vertex steps $2, 4, \ldots, 36$, from the convex 36-gon to the diameters. Each of them is a periodic path of a ball reflected inside that circle. Such a path closes because its central angle $2\pi j/m$ is a rational fraction of a full turn; with an irrational fraction it would never close.

3. The infinite equal-length original is dense in the plane

This section concerns the union over *all* depths, not the finite picture on page 1. Keep $m = 72$, $r = 1$ and $\theta = 0$.

Proposition 3. *The union of all endpoint generations is dense in* $\mathbb{R}^2$. *Consequently,*

$$\overline{\bigcup_{n\geq 1} A_n} = \mathbb{R}^2.$$

The unclosed endpoint union has Hausdorff dimension zero, the unclosed trace union has dimension one, and the closed trace union has dimension two.

Proof. Every direction has its opposite in the 72-direction set. The union $G = \bigcup_{k\geq 0} E_k$ is therefore the additive group generated by the vectors Lu_j: every integer linear combination is a finite sum using positive or opposite directions.

The directions u_0, u_6, u_{12} and u_{18} point at 0, 30, 60 and 90 degrees. In units of L,

$$2u_6 - u_{18} = (\sqrt{3}, 0),$$
$$2u_{12} - u_0 = (0, \sqrt{3}).$$

The unit axis vectors u_0 and u_{18} also occur. Thus

$$L(\mathbb{Z} + \sqrt{3}\mathbb{Z}) \times L(\mathbb{Z} + \sqrt{3}\mathbb{Z}) \subset G.$$

The subgroup $\mathbb{Z} + \sqrt{3}\mathbb{Z}$ is dense in $\mathbb{R}$. To see this directly, the pigeonhole principle applied to the fractional parts of $0, \sqrt{3}, \ldots, N\sqrt{3}$ gives a nonzero group element of magnitude less than $1/N$. It is nonzero because $\sqrt{3}$ is irrational. Changing its sign makes it positive; its integer multiples approximate any real number within that magnitude. The product of the two dense subgroups is dense in the plane.

Every endpoint belongs to a trace, so the trace closure is the plane as well. The endpoint union is countable. The trace union is a countable union of segments and contains a nondegenerate segment. Countable stability of Hausdorff dimension gives dimensions zero and one, respectively. The plane has dimension two. □

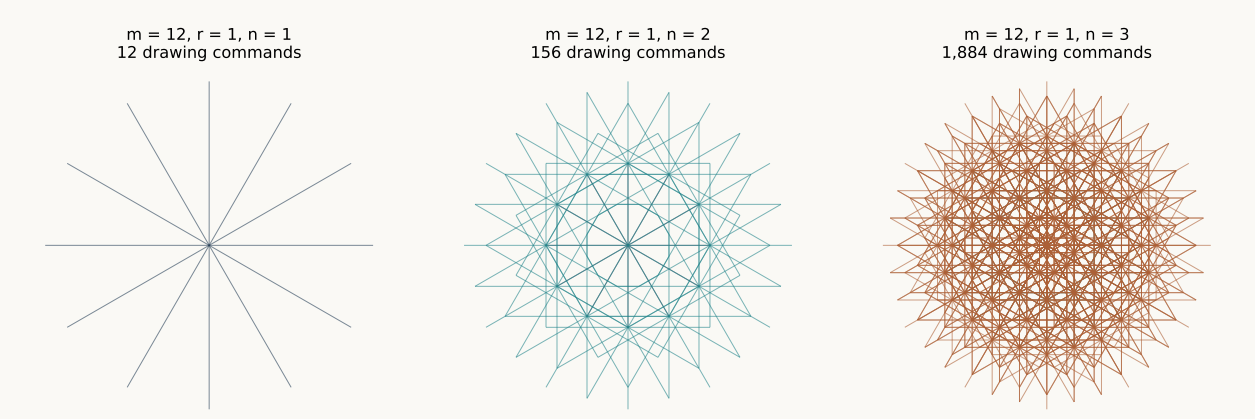


Figure 2: Three finite stages with 12 directions, a subset of the 72-direction set containing the same axis and 30-degree directions used in the proof. Each panel is scaled to its own radius. Finite images do not by themselves prove density.

The outer radius grows as nL. There is no bounded compact attractor in this equal-length case, and compact-set Hausdorff convergence to a bounded limit is not available. Density is a property of the union's closure; it does not make the unclosed countable endpoint union two-dimensional.

4. Contractive endpoints: an established polygon IFS

For $0 < r < 1$, define the infinite endpoint set

$$K = \left\{ L \sum_{t=1}^{\infty} r^{t-1} u_{j_t} : j_t \in \{0, \dots, m-1\} \right\}. \tag{5}$$

The series converges uniformly over addresses, so the address map is continuous on the compact space of addresses. Its image K is compact and satisfies $K = \bigcup_j f_j(K)$ for $f_j(x) = Lu_j + rx$; it is the unique nonempty compact attractor of these contractions, as in the standard IFS theory of Hutchinson [1].

Set $v_j = Lu_j/(1-r)$. Then

$$f_j(x) = rx + (1-r)v_j = v_j + r(x - v_j). \tag{6}$$

These are precisely contractions about vertices of a regular polygon. Polygon-vertex systems, polygonal seeds and star-shaped seeds already appear in the literature, for example Tzanov [2]. This change of coordinates and notation does not create a new endpoint-IFS family.

The convex hull of K is the regular polygon with vertices v_j. Every series is $L/(1-r)$ times a convex combination of the u_j; each vertex occurs as a constant address. Truncating and extending addresses yields the explicit approximation bound

$$d_H(E_n, K) \le \frac{Lr^n}{1-r}, \tag{7}$$

where d_H denotes the Hausdorff distance between nonempty compact sets.

A sufficient separation certificate

Each $f_j(K)$ lies in the closed disk centered at Lu_j with radius $rL/(1-r)$. Neighboring centers have distance $2L\sin(\pi/m)$. Consequently, these disks, and hence the first-level pieces, are disjoint if

$$r < q_m := \frac{\sin(\pi/m)}{1 + \sin(\pi/m)}. \tag{8}$$

This sufficient criterion is not a necessary threshold for separation. Under it, the open disk of radius $L/(1-r)$ about the origin is mapped into itself with disjoint images, giving the open set condition. The standard similarity dimension theorem [1] then gives

$$\dim_{\mathrm{H}} K = s := \frac{\log m}{\log(1/r)}. \tag{9}$$

Without a verified separation condition, only the general bound $\dim_{\mathrm{H}} K \le \min\{2, s\}$ is asserted here. For $m = 72$, $q_m \approx 0.0417962601$; the choice $r = 0.03$ therefore has the certified dimension $\dim_{\mathrm{H}} K \approx 1.219619423$. No pixel-based estimate is used.

A filled case: even m at scale $1/2$

For even m and $r = 1/2$, K is the filled regular polygon P with vertices $v_j = 2Lu_j$. Each $f_j(x) = (x + v_j)/2$ maps P into itself, so it suffices that the images cover P. Write $x \in P$ as $\alpha v_j + \beta v_{j+1}$ for some j, with $\alpha, \beta \ge 0$, $\alpha + \beta \le 1$ and, by symmetry, $\alpha \ge \beta$. Then $x = f_j(y)$ for $y = (2\alpha - 1)v_j + 2\beta v_{j+1}$. If $\alpha \ge 1/2$, y lies in the triangle $0, v_j, v_{j+1}$. Otherwise $y = (1-2\alpha)v_{j+m/2} + 2\beta v_{j+1}$, since $-v_j = v_{j+m/2}$; the coefficients are nonnegative with sum at most one, so y is a convex combination of two vertices and the center. Either way $y \in P$, so $P = \bigcup_j f_j(P)$, and uniqueness of the attractor gives $K = P$ with $\dim_{\mathrm{H}} K = 2$. For $m = 72$ the maps overlap and $s \approx 6.169925$ is no dimension at all. For odd m the point $-v_j$ lies outside P and the center is not covered.

5. The accumulated trace and its closed limit

Let $O = \bigcup_{n\geq 1} A_n$ and $S = \overline{O}$ for $0 < r < 1$. The persistent initial star generally separates the closed trace from the endpoint attractor: $S = K$ holds if and only if $B \subset K$, because every image of B under compositions of the f_j then lies in K. This fails for $m = 3$, $r = 1/2$, where the center is not a point of the Sierpinski triangle, and whenever K is totally disconnected. It holds for even m at $r = 1/2$, where K is the filled polygon of Section 4.

Proposition 4. *The closed trace is*

$$S = O \cup K = B \cup \bigcup_j f_j(S), \qquad \dim_{\mathrm{H}} S = \max\{1, \dim_{\mathrm{H}} K\}.$$

For $n \geq 1$, $d_H(A_n, S) \leq Lr^n/(1-r)$.

Proof. Every infinite address is approximated by its endpoints, so $K \subset S$. For a convergent sequence of trace points, bounded generations stay in a finite closed union of segments. For unbounded generations, the distance from a point on a generation-k segment to an infinite extension of that address is at most $Lr^{k-1}/(1-r)$, which tends to zero. Thus every remaining limit point lies in K, proving $S = O \cup K$. Taking closures in the finite recursion gives the displayed set equation. The countable segment union has dimension one. Countable stability gives the dimension formula. Points in later generations and in K lie within $Lr^n/(1-r)$ of their depth-n prefixes in A_n. □

This is a specialization of the established inhomogeneous self-similar construction with condensation set B, including its usual homogeneous-plus-orbital decomposition and Hausdorff-dimension rule [3]. These identities are not presented as new general theorems.

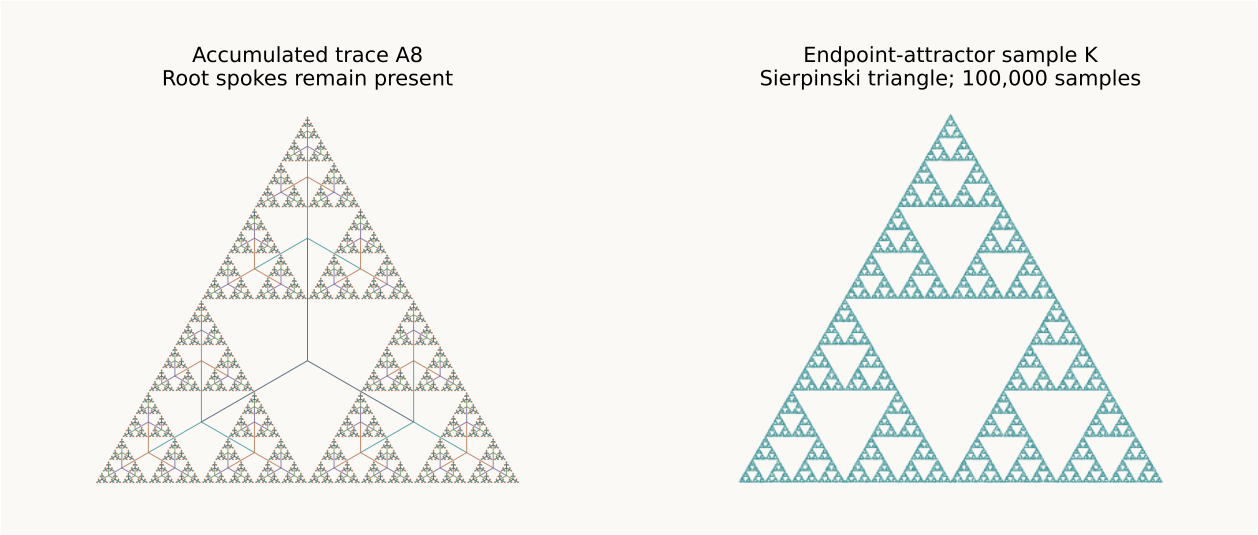


Figure 3: $m = 3$, $r = 1/2$, $L = 1$, $\theta = 90°$. Left: the trace through eight generations, including persistent spokes. Right: a finite chaos-game sample of the classical Sierpinski endpoint attractor. The two sets share the same closed-limit Hausdorff dimension, but are different sets.

An endpoint dust can have dimension below one, while its accumulated trace closure still has dimension one. For example $m = 8$, $r = 0.1$ gives $\dim_{\mathrm{H}} K = \log 8/\log 10 \approx 0.903090$ and $\dim_{\mathrm{H}} S = 1$. Connectedness of the trace and disconnectedness of many endpoint attractors are compatible with these dimension statements.

6. Representative endpoint variants

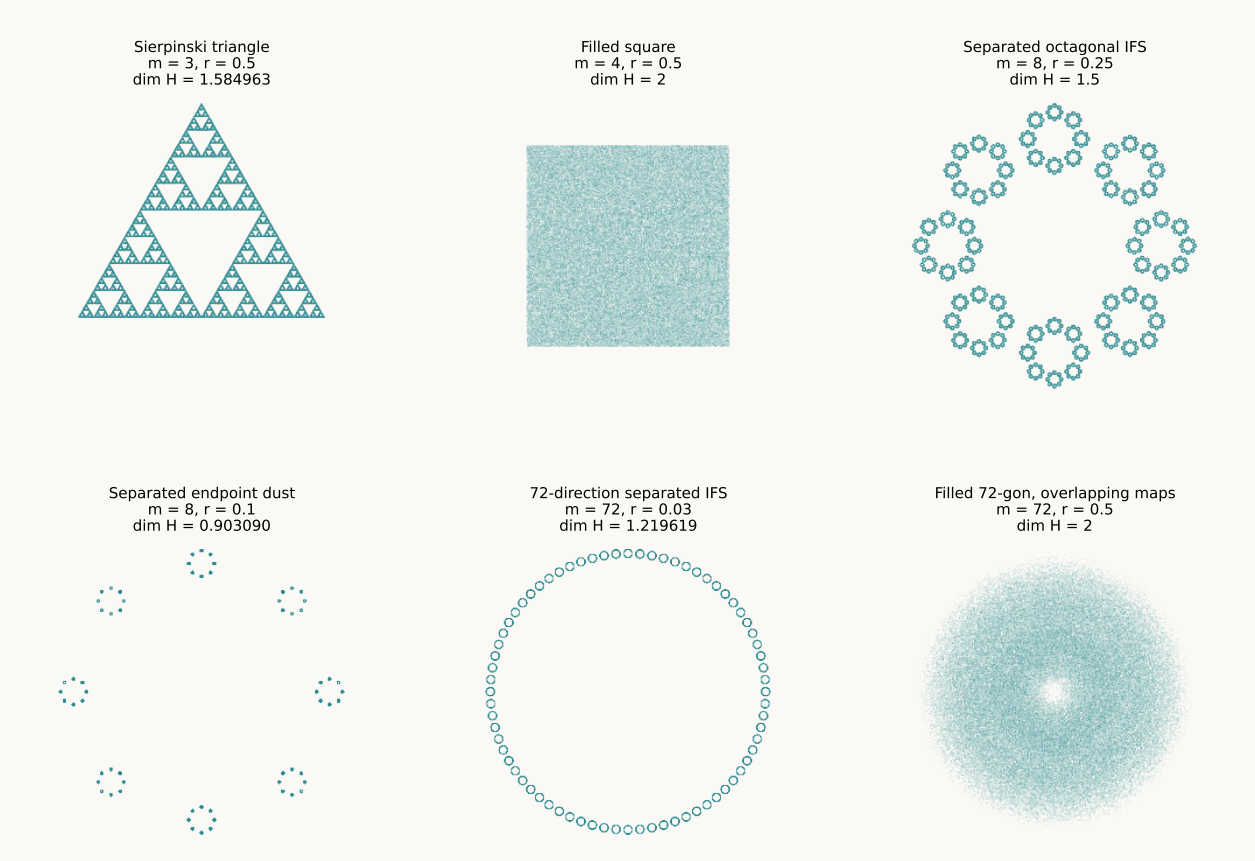


Figure 4: Each panel contains 120,000 chaos-game samples with seed 20260930; the first two panels use $\theta = 90°$ and $\theta = 45°$. These finite stochastic plots illustrate endpoint attractors, not accumulated traces. Sample density is not a picture of the set: the attractor of the last panel is the whole filled 72-gon. Dimensions below are justified analytically.

m	r	$\dim_{\mathrm{H}} K$	$\dim_{\mathrm{H}} S$	Certificate
3	0.5	1.584963	1.584963	Sierpinski triangle; OSC
4	0.5	2	2	Four squares tile the square
8	0.25	1.5	1.5	$r < q_8$
8	0.1	0.903090	1	$r < q_8$
72	0.03	1.219619	1.219619	$r < q_{72}$
72	0.5	2	2	Filled 72-gon; even m, Section 4

For the triangle at scale $1/2$, the three open corner triangles have disjoint interiors inside the original triangle. The open set condition therefore gives $\log 3/\log 2$. For the square at scale $1/2$, the four corner copies tile the entire square; uniqueness of the compact attractor makes K that square. A contractive IFS need not have a noninteger dimension or a perforated attractor. In the last panel the pale center and the thin rim are sampling effects: points there require long runs of a single direction and are rarely drawn.

Changing L rescales every set uniformly; changing θ rotates it. Neither changes these dimensions. No dimension is inferred from opacity, image resolution, repeated overdraw or apparent empty regions in the point samples.

7. Reference software and verification

The package `adler_fractal/` implements the definitions with Python floating-point coordinates. Geometry and SVG export use only the standard library. PNG output and the report figures use Matplotlib; `requirements.txt` pins Matplotlib 3.10.8 and NumPy 2.3.5. Geometry and SVG need Python 3.10 or later; the pinned figure environment needs Python 3.11 or later.

Three output meanings

`trace` Enumerates every segment command through depth n, preserving traversal order and multiplicity.

`endpoints` Enumerates depth-n endpoint addresses, with multiplicity; this is not a deduplicated point set.

`chaos` Samples $x_{t+1} = rx_t + Lu_{J_t}$ with uniform independent direction choices from a seeded generator (default seed 20260930). It requires $0 < r < 1$; n is irrelevant. The automatic burn-in bounds initial-position error relative to the limiting radius by 10^{-12}. Sampling is not an exact attractor representation.

Checks performed

All 13 tests in `tests/test_geometry.py` pass. The principal equivalence test executes the drawing function of the original script with a minimal Turtle recorder, avoiding a GUI and providing no global turtle. It compares all 5,256 segments in order with the reference output; start and end coordinates agree within 10^{-10} length units. Further tests check the explicit word formula, counts, generation lengths, sharp radii, dihedral endpoint symmetry, second-generation multiplicities, the tangent circles and chords of Proposition 2, translation, adapter behavior, empty depth, invalid parameters, SVG validity, budget rejection, seeded reproducibility and the dimension certificates. The software certifies a dimension only under the disk criterion and at scale 1/2 for the triangle and for even m; otherwise it reports none. Numerical tests support the implementation; the infinite-set claims depend on the proofs.

Reproduction

Run in the unpacked project directory:

```
python3 -m adler_fractal --output adler_original.svg
python3 -m adler_fractal --spokes 8 --scale 0.25 \
    --mode chaos --samples 100000 --output sample.svg
python3 -m unittest discover -s tests -v
python3 -m pip install -r requirements.txt
python3 scripts/make_figures.py
```

The LaTeX source builds with `tectonic docs/Adler_Fractal_Manuscript.tex`, or from `docs/` with two passes of `xelatex`. The release archive lists SHA-256 checksums of all files.

Resource and numerical limitations

Enumeration takes time proportional to the number of commands or addresses. The explicit traversal stack uses $O(mn)$ pending storage; Matplotlib additionally collects rendered data in memory. SVG is streamed. The default budget of one million rejects the depth-four original before writing an SVG; it can be raised explicitly. Float rounding and rendering choices do not change the exact mathematical definition.

8. Related work, scope and publication status

The contractive endpoint system is an established polygon-vertex IFS. The trace closure is an established inhomogeneous self-similar set. Hutchinson supplies the general compact-attractor and dimension theory [1]; Tzanov treats polygon and star-based IFS constructions [2]; Fraser states the condensation-set decomposition and Hausdorff-dimension rule [3]. The equal-length density proof and finite combinatorial calculations here provide direct analyses of the original parameters, without asserting that these observations are first discoveries.

The scoped literature check on 30 September 2026 included searches for the proposed name, radial recursive stars, equal-length spoke drawings, regular polygon IFS and inhomogeneous self-similar sets. The three listed primary full texts were examined. This was not an exhaustive review of MathSciNet, zbMATH, all books, code archives or unpublished work. The check did not locate an exact earlier instance of this finite Turtle program. Absence of a located exact instance is not evidence of historical priority.

Meaning of the proposed name

Adler Fractal is my name for the documented construction and its implementation: a descriptive proposal, not an established term. Finite drawings have dimension one; the original's infinite closure is the plane; contractive limits include dust, the classical Sierpinski triangle and filled polygons. The name does not imply that every parameter choice defines a new or noninteger-dimensional fractal. Finiteness of a rendered picture alone does not distinguish a fractal approximation from another finite drawing; the underlying limiting rule and the specified set are what is analyzed here.

Publication and licenses

This report is archived on Zenodo together with the German specification, the literature notes, the reference code and the figures; the DOI on the title page resolves to the record. The software is released under the MIT license, the text and the figures under CC BY 4.0. The exact scopes are stated in `LICENSE_DOCUMENTATION.md`. The report has not been peer reviewed. A DOI makes a deposited record citable; it is not a novelty certificate, a peer review or an official assignment of the name.

Version history

1.1.0 (1 November 2026), DOI `10.5281/zenodo.23867302`. New Proposition 2 on tangent circles and chords; the former Propositions 2 and 3 are now 3 and 4. **1.0.1** (30 September 2026), DOI `10.5281/zenodo.23063185`. Corrections to the original script, to Section 5 and to the case $m = 72$, $r = 1/2$; the definitions and the three propositions were unchanged. **1.0.0** (30 September 2026), DOI `10.5281/zenodo.23057945`. First public version.

References

[1] J. E. Hutchinson. *Fractals and Self-Similarity*. Indiana University Mathematics Journal 30 (1981), 713–747. doi:10.1512/iumj.1981.30.30055.

[2] V. Tzanov. *Strictly self-similar fractals composed of star-polygons that are attractors of Iterated Function Systems* (2015). arXiv:1502.01384.

[3] J. M. Fraser. *Inhomogeneous self-similar sets and box dimensions*. Studia Mathematica 213 (2012), 133–156. arXiv:1301.1881 (submitted 2013), especially §1.1.